

Presentation

On

EMAIL AUTOMATION USING LLMS

Group Number: 26

Presented by

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1.Introduction

- Email remains one of the most widely used communication channels globally, with over 300 billion emails sent daily across professional and personal domains.
- Manual email drafting is time-consuming, repetitive, and often lacks personalization, leading to reduced productivity in enterprise environments.
- Recent advances in Large Language Models (LLMs) offer a transformative approach to intelligent email automation, enabling context-aware generation, summarization, and personalized communication at scale.
- This project proposes an AI-powered email automation platform that leverages LLM capabilities to generate professional email drafts, summarize threads, rewrite content in different tones, classify emails, and detect priority levels.

2. Research Objectives

1. Develop an inbox-native interaction layer that captures active conversation context from webmail clients without interrupting the user workflow.

2. Design a context-preserving orchestration layer with token budgeting, preference caching, and task-aware routing.

3. Incorporate memory-aware personalization for tone control, signature reuse, and sender-specific response policy.

4. Introduce observability for latency, token consumption, estimated cost, and routing efficiency.

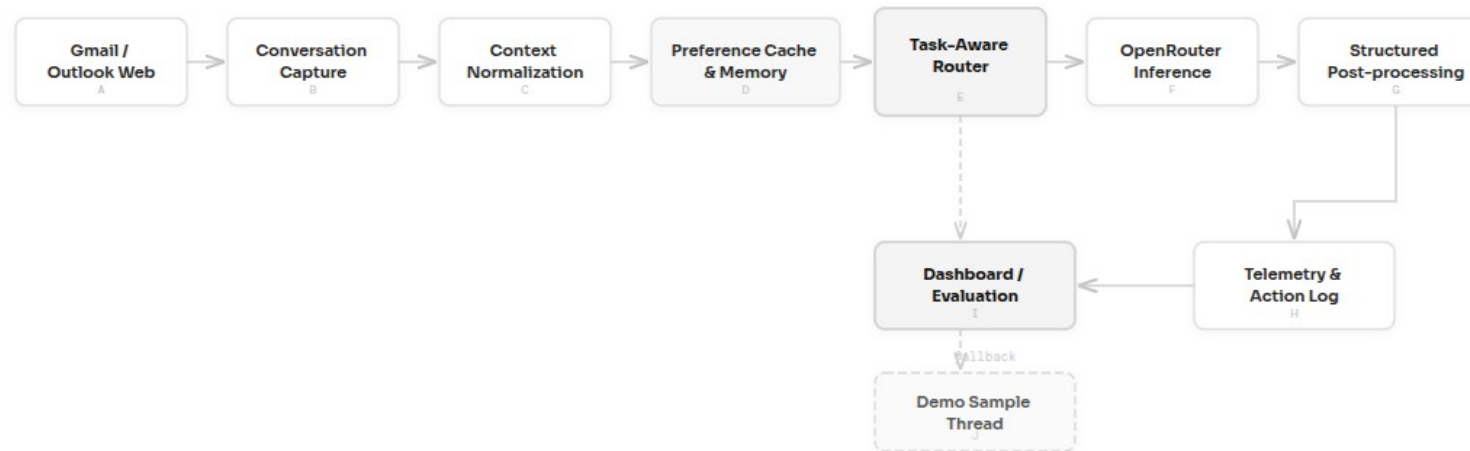
5. Ensure resilience through fallback inference, graceful degradation, and rate-limit-tolerant request handling under upstream service constraints.

3.Literature Survey

Existing Approach	Strength	Limitation	Why Our System Is Different
Smart compose systems	Fast completion	No full-thread reasoning	Handles complete conversation context
Manual LLM use	Flexible prompting	Workflow breaks on context transfer	Embedded in inbox workflow
Enterprise copilots	Strong integration	Opaque routing/cost	Explicit telemetry and routing
Template-based systems	Deterministic	Weak tone adaptation	Context-aware generation

Research gap: A browser-native, context-preserving, cost-aware, and provider-flexible email orchestration system with explicit telemetry and evaluation.

4. Proposed Model

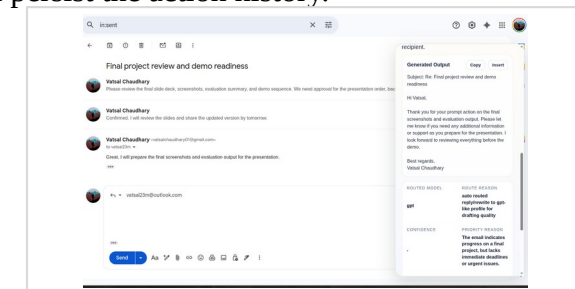


5. Discussion About Methodology & Algorithm



1. Extract the active thread and assemble a bounded conversation context.
2. Normalize the thread by removing UI noise and unnecessary fragments.
3. Resolve user memory, tone, signature, and default profile.
4. Route the task to the appropriate OpenRouter-backed profile.

5. Generate the output with task-specific prompts and context constraints.
6. Parse structured metadata such as confidence, priority, and action items.
7. Record telemetry including route reason, latency, token usage, and estimated cost.
8. Render the result in the interface and persist the action history.



6.Conclusion & Future Work

Conclusion

- The project shows that email automation becomes substantially more effective when generation is combined with thread context, model routing, and preference memory.
- The system shifts email handling from manual drafting to an intelligent orchestration workflow with visible cost and latency trade-offs.
- The architecture also improves explainability through route reason, structured metadata, and telemetry.

Future Work

Stronger Outlook parsing and layout stabilization

Attachment-aware summarization

Multilingual support

Persistent long-term memory storage

Controlled user studies and quality scoring

7. Research Paper Progress

Problem definition

Literature survey

System implementation

Evaluation tuning

Final writing

Final project review and demo readiness

Vatsal Chaudhary - vatsalchaudhary01@gmail.com
to vatsal23m

Please review the final slide deck, screenshots, evaluation summary, and demo sequence. We need approval for the presentation order, bas PM. Action items include confirming the architecture slide, validating the benchmark page, and preparing the fallback demo.

Vatsal Chaudhary
to vatsal23m

Confirmed. I will review the slides and share the updated version by tomorrow.

Vatsal Chaudhary
to vatsal23m

Great. I will prepare the final screenshots and evaluation output for the presentation.

Reply Forward

Generated Output Copy Insert

Subject: Presentation Preparation - Urgent Review Required

Summary:
The email thread discusses final preparations for an upcoming presentation, with a critical deadline of Thursday at 5 PM. Key action items include:

Action Items:
1. Review and approve presentation slide deck
2. Confirm architecture slide details
3. Validate benchmark page
4. Prepare fallback demo sequence
5. Finalize presentation order and formatting

Urgency: High - Deadline is Thursday at 5 PM

Next Steps:
- Collaborators are actively working on slide deck, screenshots, and evaluation materials
- Expecting final review and comprehensive approval before the deadline

Recommended Immediate Action: Conduct a comprehensive review of all presentation materials to ensure alignment and readiness.

Memory and Personalization
Set profile, tone, signature, and instructions used in replies

Current Memory Profile
Personalized based on AI's explicit memory and applied during AI generation

USER NAME	RECALL MODEL
Vatsal Chaudhary	AIPTD

PREFERRED TONE	CLIENT TONE
Professional	Warm And Consultative

INTERNAL TONE	EXTERNAL TONE
Direct And Concise	Extroverted Best regards, Vatsal Chaudhary

APPLY PERSONALIZATION EFFECT
Reply generation uses default profile **AIPTD**, tone fallback **professional**, under-aware tone switching (clients): **warm and consultative**, internal: **direct and concise**, your signature, and custom instructions below.

CUSTOM INSTRUCTIONS
Prioritize clarity, include clear next steps, and keep reply drafts under 140 words unless the email is strategic.

Updated Apr 3, 3:01 PM

Update Memory Settings
Customize signature and default behavior

User Name: Vatsal Chaudhary
Default Model: Auto (OpenRouter Smart Routing)

Preferred Tone: professional
Client Tone: warm and consultative

Internal Tone: direct and concise
Signature: Best regards, Vatsal Chaudhary

Custom Instructions:
Prioritize clarity, include clear next steps, and keep reply drafts under 140 words unless the email is strategic.

Changes are sent to: 1987 (vatsal23m@openrouter.com) and are used by the extension when creating drafts.

Save

OpenRouter Routing Active

Extension Dashboard
Track OpenRouter routing settings and draft generation

Statistics

Estimated Emails	Unread Drafts	Pending Replies	AI Replies
2	17	6	22

Key Metrics

Total Tokens	Cost Used	API Latency	Routing Savings
10,189	\$0.2517	2719 ms	\$0.0539

Model Routing Mix
Visualize OpenRouter model profile usage

Action Distribution
Analyze reply, routing, and reply volume

Bar chart showing Action Distribution: reply, routing, and reply volume.

8.References

1. Vaswani, A. et al. Attention Is All You Need. NeurIPS, 2017.

2. Lewis, M. et al. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. ACL, 2020.

3. Google. Gmail API Documentation. <https://developers.google.com/gmail>

4. Microsoft. Microsoft Graph and Outlook Add-in Documentation. <https://learn.microsoft.com/graph>

5. OpenRouter. API Documentation and Model Routing Reference. <https://openrouter.ai/docs>